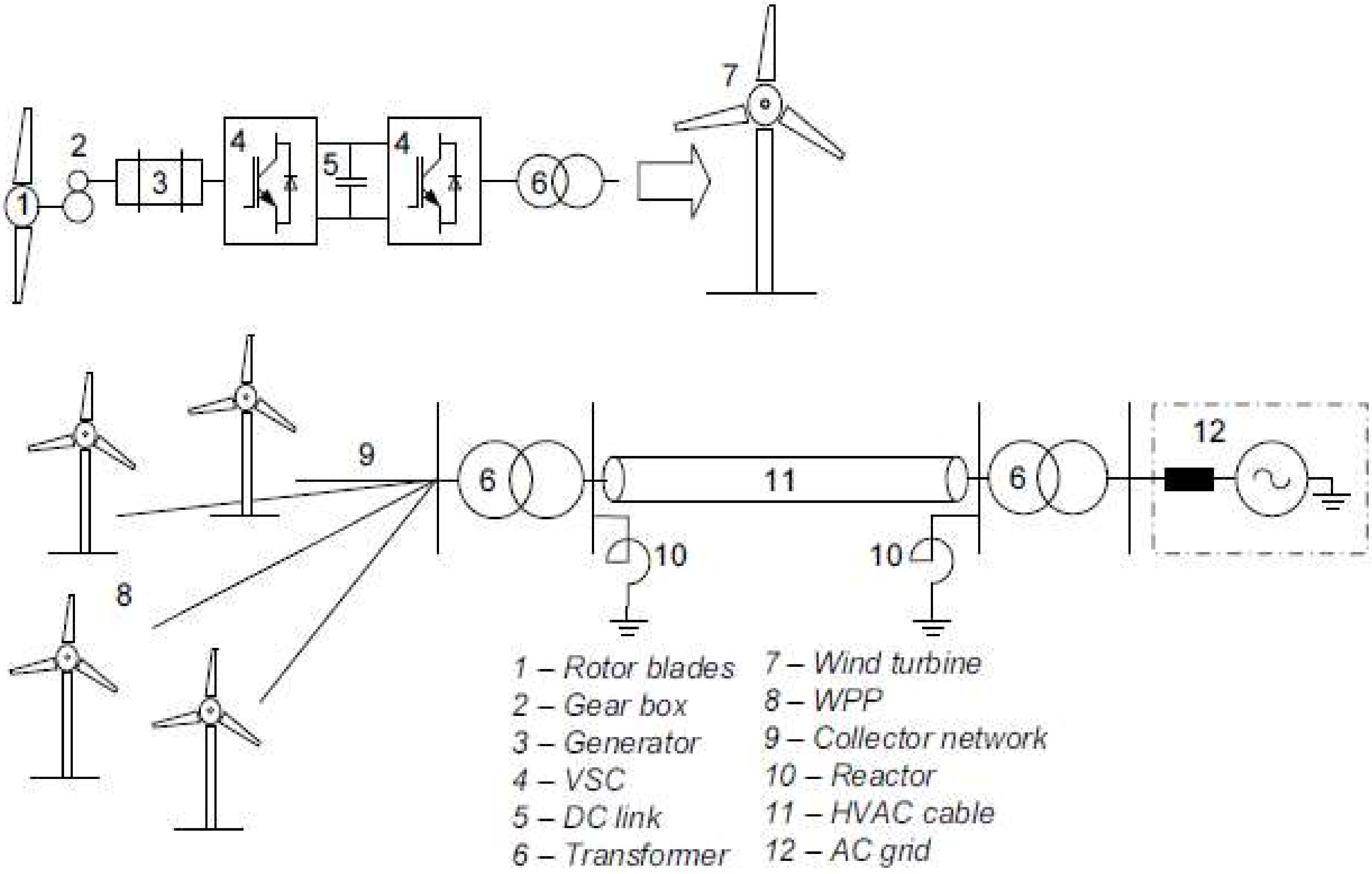
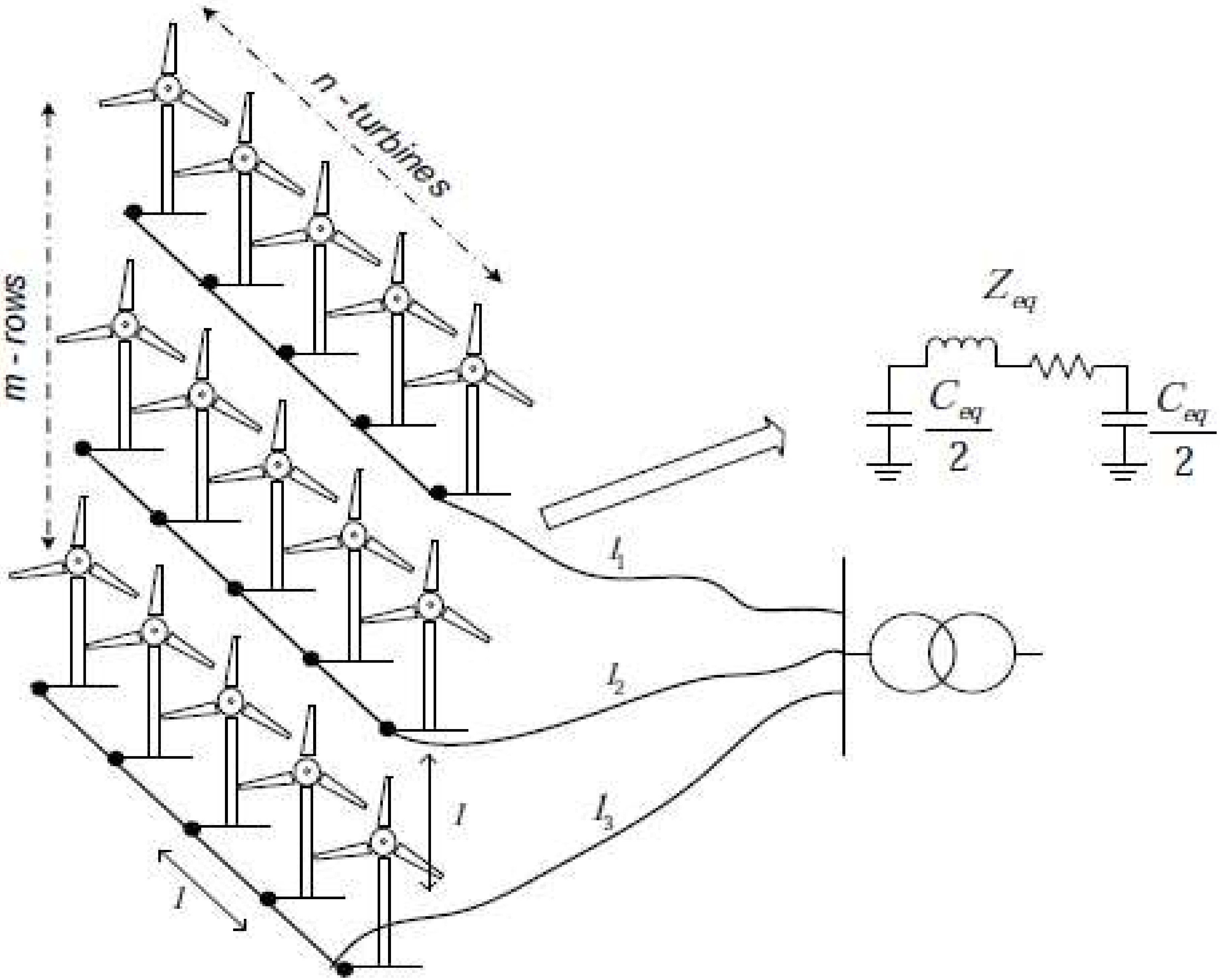
Wind Turbines

Structure of Wind Power Plant : HVAC



Layout of A Wind Power Plant



- A wind turbine is a rotating machine which converts the kinetic energy of wind into mechanical energy.
- If the mechanical energy is used directly by machinery, such as a pump or grinding stones, the machine is usually called a windmill.
- If the mechanical energy is instead converted to electricity, the machine is called a wind generator, wind turbine, wind power unit (WPU), wind energy converter (WEC), or aerogenerator.
- Horizontal Axis Wind Turbines (HAWT)
- Vertical Axis Wind Turbines (VAWT)

Installed wind power capacity (MW) and share of total electricity consumption

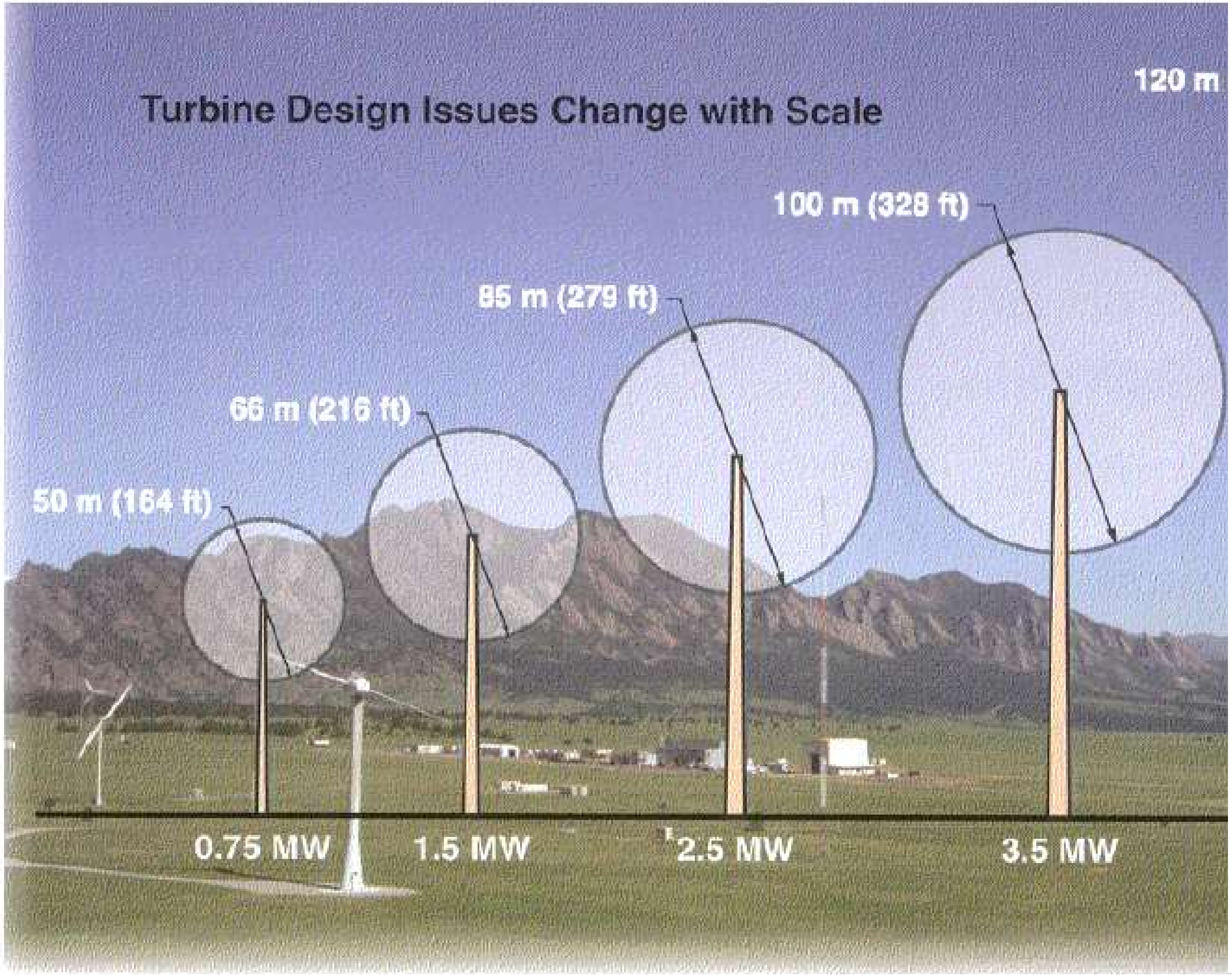
#	Country or territory	2014[14]	2015[3]	2016[15]	2017[16]	2018[17]	2019[13]	2020[18]	2021[19]	2022[20]	Per capita (watts) (2020)	Share of total consumption
1	China	114,763	145,104	168,690	188,232	211,392	236,320	281,993	328,973	365,964	200	6.1% (2020)
–	European Union	128,752	141,579	153,730	169,319	178,826	192,020	201,507	187,497*	202,700*		
2	United States	65,879	74,472	82,183	89,077	96,665	105,466	117,744	132,738	140,862	355	9.2% (2021)
3	Germany	39,165	44,947	50,019	56,132	59,311	61,357	62,184	63,760	66,315	747	23.1% (2021)
4	India	22,465	27,151	28,665	32,848	35,129	37,506	38,559	40,067	41,930	28	
5	Spain	22,987	23,025	23,075	23,170	23,494	25,808	27,089	27,497	29,308	572	19% (2015)
6	United Kingdom	12,440	13,603	15,030	18,872	20,970	23,515	24,665	27,130	28,537	369	21% (2021)
7	Brazil	5,939	8,715	10,740	12,763	14,707	15,452	17,198	21,161	24,163	81	
8	France	9,285	10,358	12,065	13,759	15,309	16,643	17,382	18,676	21,120	258	4.3% (2015)
9	Canada[21]	9,694	11,205	11,898	12,239	12,816	13,413	13,577	14,304	15,295	353	
10	Sweden	5,425	6,025	6,519	6,691	7,407	8,804	9,688	12,080	14,557	933	20.3% (2020)
11	Italy	8,663	8,958	9,257	9,479	9,958	10,512	10,839	11,276	11,780	183	7.1% (2019)
12	Turkey[22]	3,763	4,718	6,101	6,516	7,369	8,056	8,832	10,607	11,396	106	
13	Australia[23]	3,806	4,187	4,327	4,557	5,362	6,199	8,603	8,951	10,134	367	
14	Netherlands	2,805	3,431	4,328	4,341	4,471	4,600	6,600	7,801	8,688	375	

Vestas 8 MW WIND TURBINE

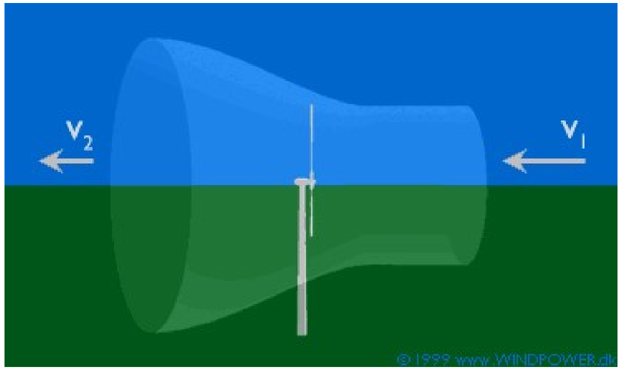
- Tower height: 140-meters
- Tip height: 220 meters
- Swept area: 21,000

GE Wind Energy	Haliade-X	13	Offshore	Commercially deployed	August 2023	Dogger Bank A, Dogger Bank B (United Kingdom)	[26]
Siemens Gamesa	SG 11.0-200 DD	11	Offshore	Commercially deployed	April 2022	Hollandse Kust Zuid (Netherlands), Hollandse Kust Noord (Netherlands)	[27]
Mingyang Wind Power	MySE 11-203	11	Offshore	Concept			Prototype planned in 2021 with commercial availability in 2022.[28][29]
Dongfang Electric	D10000-185	10	Offshore	Prototype	July 2020	Xinghua Bay (China)	Prototype deployed July 2020. Rotor diameter of 185 meters (607 feet). [30][31]
MHI-Vestas	V164-10.0	10	Offshore	Upgrade of current model		Seagreen (United Kingdom)	Incremental upgrade from currently deployed V164 model[32]
MHI-Vestas	V164-9.5	9.5	Offshore	Commercially deployed	December 2019	Northwester 2 (Belgium) Kincardine (Aberdeen, United Kingdom)	[33][34]
Enercon	Enercon EN-220/10MW	10	Onshore	World's largest onshore wind turbine [35]		China's northern region	
Enercon	E-126 7.580	7.5	Onshore	No longer offered for sale[36]		Magdeburg-Rothensee, Ellern (Germany), Estinnes, (Belgium), Noordoostpolder (Netherlands)	Most powerful onshore wind turbine ever offered for sale[37][38]

Reality of Capacity Vs Size



BETS LAW



Let V_1 be the velocity of the wind intercepted by the wind turbine and let V_2 be the velocity of the wind after it passes the WT.

Let S be the swept area of the turbine(The area swept by the Turbine blades) and ρ be the density of air.

Average wind speed =$(V_1 + V_2)/2$

Mass flow rate of air "m" = $\rho S(V_1 + V_2)/2$

Power extracted from the Wind by the

P_0 be the Power contained in the wind if not intercepted by the Turbine

$P_{0=\rho S V_1^2}/2$------Eq(2)

Coefficient of Power Extraction $C_p = P/P_0$

$C_p = (1-(V_2/V_1)^2)(1+V_2/V_1)$

Let $b = V_2/V_1$, $C_p = (1-b^2)(1+b)$----Eq(3)

For C_p to be maximum $dC_p/db = 0$

This can be done based on

$d/dx(u.v) = u\, dv/dx + v dv/dx$

differentiating, $\{(1-b^2)1 + (1+b)(-2b)\}\ /2 = 0$

$= \{(1-b^2)-2b-2b^2)\}\ /2 = 0$

$= \{(1-3b-3b^2)\}\ /2=0$

$=(1-3b)(1+b)=0$

$b=1/3$ or $b=-1$

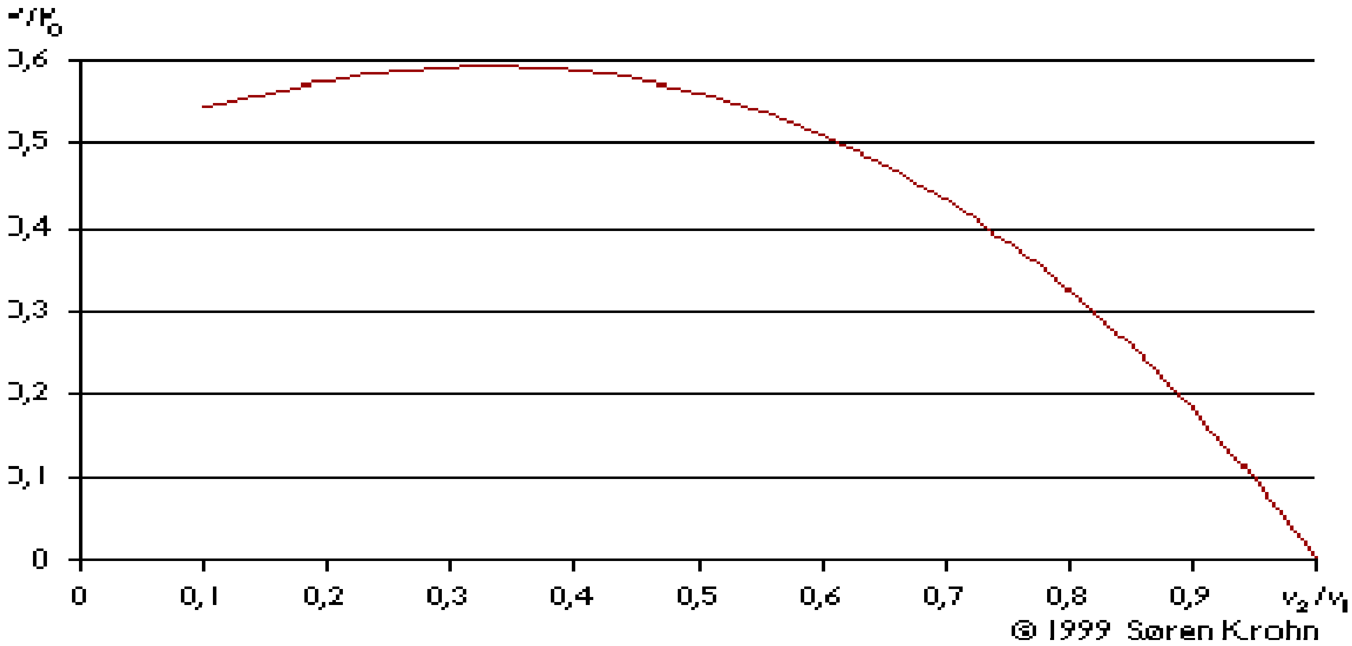
Hence for maximum Power extraction $b=1/3$ ie $V_2/V_1=1/3$

Hence Maximum possible $C_p = (1-(1/3)^2(1+1/3)$

$=16/27$

$=59.26\%$

We may plot P/P_0 as a function of v_2/v_1:



The Power Extraction Analysis